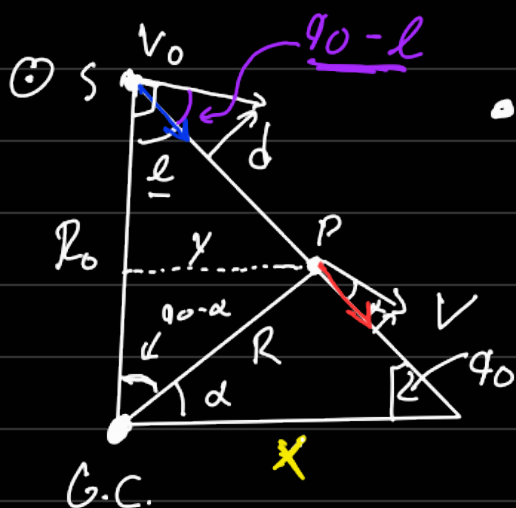




• Vectors are additive

$$\begin{aligned}
 V_r &= (V_0)_r + (V)_r \\
 &= V_0 \cos(q_0 - l) + \cos(\alpha) V \quad * \\
 &= -V_0 \sin(l) + \cos(\alpha) V
 \end{aligned}$$

G.C.

• Use geometry of the problem to replace $\sin(l)$, $\cos(\alpha)$

$$\cos(\alpha) = \frac{x}{R} \quad \Bigg| \quad \cos(l) = \frac{x}{R_0}$$

$$\sin(q_0 - \alpha) = x/R = \cos(\alpha)$$

$$x = R \cos(\alpha) = R_0 \sin(l)$$

$$R \cos(\alpha) = R_0 \sin(l)$$

$$\cos(\alpha) = \frac{R_0}{R} \sin(l)$$

$$\begin{aligned}
 V_r &= -V_0 \sin(l) + V \left(\frac{R_0}{R} \sin(l) \right) \\
 &= \sin(l) \left(-V_0 + \frac{V R_0}{R} \right) \\
 &= \left(\frac{V R_0}{R} - V_0 \right) \sin(l)
 \end{aligned}$$

$$V_r = R_0 \left(\frac{V}{R} - \frac{V_0}{R_0} \right) \sin(l)$$

$$V^2 = V_r^2 + V_t^2 \Rightarrow V_t = \sqrt{V^2 - V_r^2}$$

$$V_t = \left(\frac{V}{R} - \frac{V_0}{R_0} \right) R_0 \cos(l) - V \frac{d}{R}$$

* Introduce the angular vel.:

$$\omega = \frac{V}{R}$$

$$\omega_0 = \frac{V_0}{R_0}$$

⊙

$$V_r = (\omega - \omega_0) R_0 \sin(l)$$

$$V_t = (\omega - \omega_0) R_0 \cos(l) - V \frac{d}{R}$$

$\omega(R)$: look at this for $R, R_0 \gg d$

$$\omega(R) = \omega(R_0) + \frac{d\omega}{dR}(R-R_0) \quad (\text{Taylor exp})$$

$$\underbrace{(\omega - \omega_0)}_{\text{Replace}} = \underbrace{\frac{d\omega}{dR}}_{\text{Replace}} (R - R_0)$$

Replace

Define Oort's Constants: A, B

$$A = - \frac{R_0}{2} \left. \frac{d\omega}{dR} \right|_{R_0} = \frac{1}{2} \left(\omega_0 - \left. \frac{dV}{dR} \right|_{R_0} \right)$$

$$B = -\frac{1}{2} \left(\frac{V_0}{R_0} + \left. \frac{dV}{dR} \right|_{R_0} \right)$$

A.2 Exam

$$A = \frac{1}{2} \left(\frac{V_0}{R_0} - \left. \frac{dV}{dR} \right|_{R_0} \right)$$

$$B = -\frac{1}{2} \left(\frac{V_0}{R_0} + \left. \frac{dV}{dR} \right|_{R_0} \right)$$

$$A = 18.0 \text{ km s}^{-1} \text{ kpc}^{-1}$$

$$B = -10.5 \text{ km s}^{-1} \text{ kpc}^{-1}$$

R_0, V_0 : 
(Sun)

• Sun: ang. vel. $\omega_0 = \frac{V_0}{R_0}$

$$\begin{aligned} A - B &= \frac{1}{2} \left(\frac{V_0}{R_0} - \frac{dV}{dR} \right) - \left[-\frac{1}{2} \left(\frac{V_0}{R_0} + \frac{dV}{dR} \right) \right] \\ &= \frac{1}{2} \left(\frac{V_0}{R_0} - \frac{dV}{dR} \right) + \frac{1}{2} \left(\frac{V_0}{R_0} + \frac{dV}{dR} \right) \end{aligned}$$

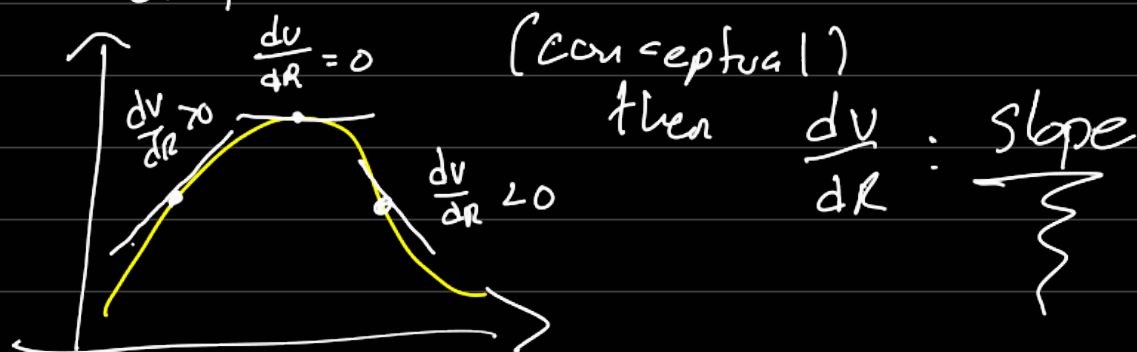
$$A - B = \frac{V_0}{R_0} = \omega_0$$

$$\omega_0 = 18 + 10 = 28.5 \quad (\text{Sun})$$

$$A + B = - \left. \frac{dV}{dR} \right|_{R_0}$$

$$\frac{dV}{dR} = -(18 - 10) = \underline{\underline{-8}}$$

1b) $V(R)$:



$\left. \frac{dV}{dR} \right|_{R_0}$: local gradient near the sun is negative meaning it is decreasing/declining

PS2 (4) | $A = 15, B = -12$

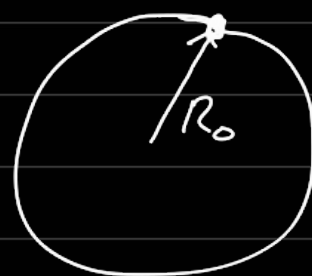
$$A - B = \omega_0 = \frac{V_0}{R_0} = \underline{\underline{\omega_0}}$$

$$\bullet X = Vt \Rightarrow \frac{X}{t} = V$$

$$"X" = 2\pi R_0$$

T : period

"V": angular velocity

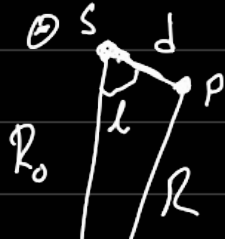


$$\underline{\omega_0} = \frac{2\pi R_0}{T} \Rightarrow \underline{\underline{T = \frac{2\pi R_0}{\omega_0}}} \quad (\text{period})$$

PS3 (1) | $V_L = (\omega - \omega_0) R_0 \cos(\ell) - \omega d$

$$V^2 = V_t^2 + V_r^2$$

Assumption $\underline{\underline{d \ll R_0}}$



$$V_t \approx d (A \cos(2\ell) + B)$$

• Main idea comes from the Taylor exp:

$$\cos \theta \approx 1 - \frac{\theta^2}{2} + \dots$$

$$\omega \approx \omega_0 + (R - R_0) \frac{d\omega}{dR} \bigg|_{R_0}$$

$$(\omega - \omega_0) = (R - R_0) \frac{d\omega}{dR}$$

G.C. $V_t = (R - R_0) \frac{d\omega}{dR} R_0 \cos(\ell) - \omega d$



$$\cos(\ell) = \frac{x'}{d}$$

$$(R - R_0) \approx d \cos(\ell)$$

• Do algebra to get in terms of A and B

$$V_t = d \cos(\ell) \left(\frac{d\omega}{dR} \right) R_0 \cos(\ell) - \omega d$$

$$= d \left(\frac{d\omega}{dR} \right) \cos^2(\ell) - \omega d$$

$$\cos^2(\ell) = \frac{1 + \cos(2\ell)}{2}$$

LSR: $V_{\text{rsm}}:$

(R, θ, h)



$$z_0 \sim z_\odot$$

